

Thinking Note

AI-native trading research platform

Question → Hypothesis → Experiment → Evidence → Learning (QHEEL model)

“Does buying NIFTY after a sharp fall work?”

1. My questions :

- > which NIFTY?
- > what does sharp fall mean?
- > what does work mean?

2. My assumptions :

- > NIFTY 50 index (most widely used when talking about nifty)
- > work meaning profit of any kind

3. What i would ask the user :

- > what is the threshold of profit to lie in work
- > how many days do you want to hold
- > calendar days or session

4. What would the experiment look like :

- > Market / instrument : from the above question i would assume NIFTY 50 until otherwise specified by the user
- > Entry condition : i would assume or ask the user the meaning of “sharp fall”
- > Exit condition : i would ask the user what does success exactly mean and define a exit condition
- > Holding period : i would ask the user how many days does he want to hold them
- > Test period : i would ask the user
- > Important costs or assumptions : i would assume a cost and then give the user option to change it

5. What could go wrong? Think about how the system could produce a misleading or incorrect conclusion. Consider issues such as:

Ambiguous definitions > for Ambiguous definitions i would either assume or ask the user to specify while showing him the assumptions i made

Incorrect assumptions > the user has the autonomy to change the assumptions in case of wrong taken initially

Data quality > data quality needs proper feature engineering and data cleaning to make sure it up to par

Look-ahead bias > i would make sure to make the backtest so that it only looks at the previous data only

Transaction costs > initially i would assume a fixed cost for transactions but it needs to be dynamic for a proper backtest

Slippage > if we would were to work on real world we can gracefully handle it by either calculating slippage using flags and properly show the user slippage as well

Overfitting > feature engineering can handle overfitting to some extent we can also perform standard procedures like reducing the parameters of the model

Insufficient evidence > there will be a many instances that there will be a lack of evidence for that we need to properly inform the user that the data in not sufficient to make a proper conclusion but still show them the data